Avo Carbon INDIA	8D REPORT	Format no:F/QAD/8D/09 Rev no:01 Rev date:15.09.2023

D0 Problem Awareness

8D-no: 2 **Submit Date:** 10.12.2024

Status: Open

Problem: Pigtail position (Insertion) NG **Supplier:** Avo Carbon India

Part No: MA001 -43019 Avo Part # 9375-504 **Place:** Chennai

Reported Date: 22.11.2024

Editor: Saravanan.B **Supplier Agent:** P. Ramkumar

D1 Establish Team

Member		Phone	E-Mail	Department
1	N.Gayathri	+91 444 2926571	gayathti.n@avocarbon.com	Head QAD
2	Suresh.P	+91 444 2926541	suresh.p@avocarbon.com	Head Production
3	Sundaresan.R	+91 444 2926525	sundaresan.r@avocarbon.com	Head Engg
4	Saravanan.B	+91 444 2926573	Saravanan.b@avocarbon.com	Asst manager-QAD
5	Suresh.A	+91 444 2926573	suresh.a@avocarbon.com	Engineer -QAD

D2 Problem Description

Problem no: 2 **Quantity**

Problem Date: 22.11.2024 **Tested Quantity**

Manufacturing Date/ Month: **Failure Quantity** 1

Repeated Failure: [ ] Yes [x] No

Compliant Text: **Pigtail position (Insertion) NG**

D3 -Interim Containment Action

Title		Responsible	Plan End
1	**Sorters, Operators, Inspectors & Supervisors are alerted about the compliant. And displayed the signed copy of Q-Alert with respective area.**	A.Suresh	22.11.2024
2	**Available FG Parts 560 no's containment initiated 100% Checked found 52 numbers more than specified limit of 2.5mm**	A.Senthil	27.11.2024
3	**Around 21100 numbers in WIP -yet to do containment action**	A.Senthil	

D4 Root Causes Analysis Why- Why Analysis

Root cause-Occurrence	Responsible	Plan End
Why ? Flexible Insertion Depth High Why ? Over hanging Length High Why ? Wrong Setting Why ? Based on insertion depth over hang length will be fixed Why ? **Cause:** Flexible over hanging length was set excess length during setting due wire hanging length was fixed based on flexible insertion depth so it caused flexible insertion depth high.	Mr.Suresh	04.12.2024

Root cause-Escape	Responsible	Plan End
Why ? Pigtail Insertion NG escaped to customer Why ? Not Detected at Final inspection Why ? Acceptable and rejected limit not defined in the visual standard Why ? Why ? **Cause:** Pigtail insertion flow out due to acceptable limit was not defined in the visual standard.	Mr.Saravanan	03.12.2024

D5 Systemic Corrective Action / long term Action

Title	Responsible	Plan End
Flexible over hang length during setting was fixed 3 to 5mm. and over hanging measurement mechanism introduced.after setting length measured by measurement scale.. insertion depth cheked during first off,Insertion pattern check added in the inspection record. Visual Standard to be updated with insertion acceptable and rejected pattern. Magnified camera inspection to be introduced to visualize the defect better.	Mr.Suresh	12-Dec-24

D6 Effectiveness of implemented Corrective Action

Title	Responsible	Plan End
Insertion depth and pattern was found ok, pull strength checked meet the drawing specification 1 kgf min. actual measured values 1.5 kgf measured, insertion pattern not more than 2.5mm towards contact face.	Mr.Suresh /Mr.Saravanan	06.12.2024

Relevant Core Tools update

Title	Responsible	Plan End
Control plan, FMEA, SOP,Inspection records and change management & validation report.	Mr.Suresh.A	09.12.2024

D7 Horizontal Deployment Action

Title	Responsible	Plan End
Based on Effectiveness actions to be plan and implement to other applicable similar parts.	Mr.Suresh	15.12.2024

Preventive measure

Title	Responsible	Plan End
Flexible over hang length during setting was fixed 3 to 5mm. and over hanging measurement mechanism introduced.after setting length measured by measurement scale.. insertion depth cheked during first off,Insertion pattern check added in the inspection record. Visual Standard to be updated with insertion acceptable and rejected pattern. Magnified camera inspection to be introduced to visualize the defect better.	Mr. Suresh	12.12.2024

D8 Problem Closure

Title	Responsible	Plan End
Team Congratulations	Ms.Gayathri	12.12.2024

Revision history
01/00-16.08.2022 fresh issue
01/02-15.09.2023 overall Reviewed & Updated

Fishbone Diagram

MA001 -43019 **Defect name :** **Pigtail position (Insertion) NG** **Date :** **22.11.2024**

Customer Name : **M/s. Focustek**

Real Cause : **Flexible over hanging length excess in setting**

Man

Un trained operator

Flexible wire over hanging length Wrong setting

Setting brush mixup

Machine

Spool motor not working

The spool wire does not move properly in the ring washer

Sharp edge found on the Bottom tool

Blunt cutter tip

Upper tool hole sizes were in high clearances

Mix powder more deposited between the cutters

Tonnage low

Wire Clamping /Declamping Pressure less

Wire Clamping /Declamping Pressure high

Materials

Colour change of wire

Flexible dia over size

Flex was hard

Flex was more contaminated

Improper flex winding inside the spool

Pigtail position (Insertion) NG

Method

Flexible cutting direction

Cutting Position

Flexible Clamping distance high

Environment

Measurement

Inspection standard insertion pattern not covered

AVO Carbon INDIA	VERIFICATION AND VALIDATION	Formate# F/QAD/VAV/039 Rev# 00 Rev Date :20.03.2023

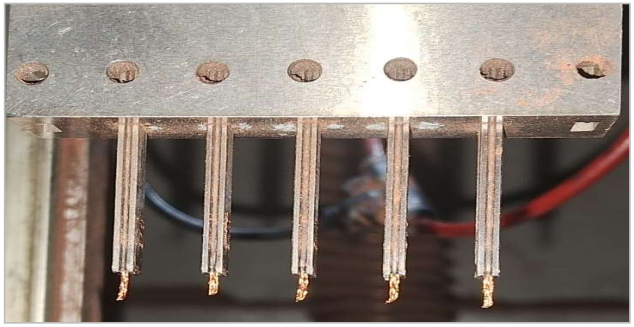
4M	S.No	Possible Causes	Occurrence-O/ Detection-D	Verification and Validation	Result
Machine	1	Spool motor not working	O	Switching any spool motor from ON to OFF in the spool selection option in the HMI will automatically stop the machine with a fault message. Fucntion of spool motor verified found ok.	OK
	2	The spool wire does not move properly in the ring washer	O	We made sure that the spool wire moves properly in the ring washer. If get wires strucks ,the machine will stop automatically will come with fault message on the HMI monitor, Wire flow verified from the spool to compression insert, mooth flow ensured.	OK
	3	Sharp edge found on the Bottom tool	O	No sharp edge found on the Bottom tool. Because the tool relief angle is given as default in all compacting tools	OK
	4	Blunt cutter tip	D	Cutter history card follows , wire cutting found no issue.	OK
	5	Upper tool hole sizes were in high clearances	D	Before loading the top tool into the machine, the wire is manually inserted into the tool hole and checked by the setter and QA.We made sure that the wire entered the tool hole smoothly.	OK
	6	Mix powder more deposited between the cutters	O	After removing the cutters, we found that no powder was deposited. The cutter screw is properly tightened with its spring tension so that powder does not settle between the cutters, ensuring that they are always within easy clearance for cutting.	OK
	7	Tonnage low	O	Brush compacting is tonage spec: 2000 DaN to 3000 DaN for 5 Cavity of tool is running. Density was achieved and we confimed that the machine actual is running at 2450DaN to 2650 DaN	OK
	8	Wire Clamping pressure less /high	O	Wire clamping pressure checked, operate as per defined specification, actual clapming pessure measured 4.5 bar.	Ok
	9	Wire De clamping pressure less /High	O	Wire de clamping pressure checked, operate as per defined specification, actual clapming pessure measured 1.5 bar.	Ok
Method	1	Flexible cutting direction	O	Cutting Direction we found one direction this was regular one of the press cutter used.	OK
	2	Cutting Position	O	Cutting position we found all pilling machine for same horizontal position found this was as desigined.	OK
	3	Flexible Clamping distance high	O	Flexible Clamping to over hanging length measured and clamping of flexible verified found ok.	Ok
Man	1	Untrained operator	O	Operator Skill Matrix found OK	OK
	2	Setting brush mixup	O	No setting brsuh Mixup found _ Setting brush handling procedure to know all awre of	OK
	3	Flexible wire over hanging length Wrong setting	O	Over hang length High	Insertion depth measures 2.50/2.75mm , During setting there is a possibility of incraese the wire hang length. There is no specification fixed for wire hang length which is indirectly measured with the part.
Material	1	Colour change of wire	O	No Colour change found /Wire Receiving stage is checked....	OK
	2	Flexible dia /construction over size	O	Receiving inspection on place wire inspected contruction whiel Receiving stage is checked, checked wire report found ok.	OK
	3	Flex was hard	O	Wire was correct elangation strength found / measured 13% against 10% (min) - Supplier report verified.	OK
	4	Flex was more contaminated	O	No wire contamination found / Wire Receiving stage is checked.	OK
	5	Improper flex winding inside the spool	O	No improper Flex winding of the spool /Wire Receiving stage is checked	OK
Measurement	1	Inspection standard insertion pattern not covered	D		
Environment	1				
	2				
	3				

Fishbone Diagram © 2009 Vertex42 LLC

Improvement - Before and After

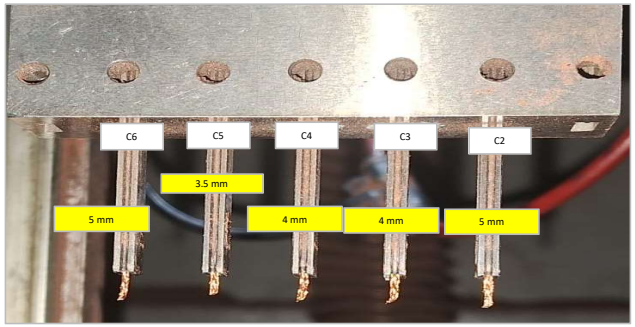
Before

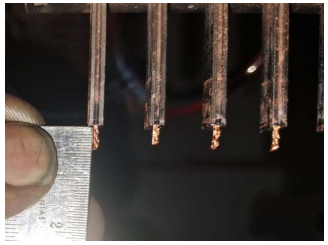
Customer Name : M/s. Focustek

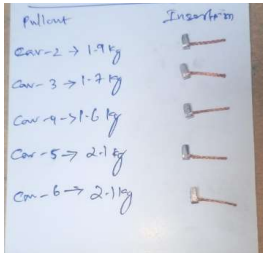


Flexible Wire over hanging length not defined, Based on insertion depth over hang length will be fixed

After







Flexible over hang length during setting was fixed 3 to 5mm. and over hanging measurement mechanism introduced.after setting length measured by measurement scale.. Insertion pattern also checked.